Programmed Radial Registry in Layered Conductive Materials

Geometric Superperiods, Interlayer Configuration, Spatial Compression, and a Falsifiable Protocol for Configuration-Dependent Transduction

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Publication edition R6 - bounded identifiability and inspectability revision rebuilt directly from R5. Retains the R5 mathematical and mechanism claim ceiling while separating measured from nominal arithmetic descriptors, bounding tooling aliasing, balancing held-out loss across states, and making the targeted prior-art search inspectable.

Abstract

This paper proposes a deliberately low-assumption experimental architecture for asking whether relative interlayer geometry carries reproducible predictive information in layered conductive materials beyond ordinary bulk deformation and contact variables. The central construction is a stack of thin conductive or conductive-composite sheets whose surfaces contain prescribed radial spatial modes around a common central actuation region. The modes are treated as manufactured geometric relations rather than as electronic moiré states. Their relative wavelengths, phases, offsets, separations, and contact topology define a configuration-dependent registry field that can be traversed continuously by central displacement.

The mathematical program derives registry superperiods, phase-translation rules, metric and curvature mismatch, contact-threshold loci, scaling invariants, hierarchical multilayer recurrences, and exact arithmetic correspondences for integer-indexed spatial modes. In particular, on a carrier of active radial length L, manufactured modes with wavelengths $\lambda_n=L/n$ and $\lambda_m=L/m$ generate a difference-mode registry scale $L/|n-m|$; their common partition has gcd(n,m) intervals, whereas their coarsest common uniform refinement has lcm(n,m) intervals. These are exact statements about the constructed geometry, not claims about microscopic mechanism.

A nested model ladder then separates bulk deformation (M0), contact/separation/network effects (M1), registry geometry (M2), material-specific interlayer effects (M3), and later mechanism-specific models (M4,j). The fabrication screen contains seven geometry classes - six modulated classes plus one unmodulated matched-roughness control - crossed with two applied-preload conditions, for 14 experimental states in total. The 12 modulated states form the common specimen population for the M0/M1/M2 held-out predictive comparison; the two U states are reserved for the separate structured-surface control in Section 16. The strongest initial positive claim is intentionally limited: deliberately prescribed interlayer registry adds reproducible predictive information under the tested conditions. No magic-angle, quantum-interference, topological-band, charge-transfer, or other microscopic claim is promoted without independent discriminating evidence.

1. Research question and evidentiary posture

The experimental question is narrow: does controlled interlayer registry improve prediction of a measured response after ordinary bulk deformation, environment, and contact variables have been accounted for? The question is intentionally posed before selection of a microscopic explanation.

The work extends an existing open-replication program in configured-material transduction. That program defines its first target as a reproducibly distinguishable response produced by intentional configuration changes in otherwise matched material systems, while explicitly refusing to require one privileged microscopic explanation [P1]. It also separates curvature from topology: ordinary shape change is geometry; topology enters only when connectivity, contacts, cracks, boundaries, or loops change [P1]. Part II extends the program from response into thresholding, memory, routing, feedback, and reconfiguration, again treating descriptive vocabulary as downstream of reproducible relations [P2]. Part III treats transition ordering, cross-mechanism coupling, and state-dependent topology as candidate design variables that must outperform simpler explanations in discriminating tests [P3].

The present paper retains that evidentiary structure. Geometry specifies where to look; measurement determines whether a reproducible relation exists; model discrimination determines what the observation warrants. A visually interesting pattern, a large electrical signal, or resemblance to an established exotic system is not by itself a mechanism claim.

2. Relation to established adjacent work

Several established research areas motivate the question without deciding it. Modern moiré-material research shows that relative layer misalignment, lattice mismatch, strain, pressure, and other interlayer controls can strongly modify properties in atomically thin crystalline heterostructures [1-3]. That literature establishes that interlayer registry can be a physically consequential control variable in suitable materials. It does not establish that a garage-scale chitosan/graphite or graphene-bearing radial stack reproduces the electronic mechanisms of twisted bilayer graphene.

A more directly relevant materials precedent is the demonstrated strain sensitivity of chitosan-based films containing few-layer graphene, where graphene content and processing influence electrical response [4]. Measured piezoelectric response has been reported in solution-cast chitosan films with strong dependence on processing and crystalline content [5]. A separate 2026 study deliberately uses the more cautious term piezoelectric-like and shows that hydration, substrate, and casting conditions materially alter the measured chitosan response [6]. Curvature-associated flexoelectric behavior in graphene nanowrinkles has now been published in peer-reviewed form [7]. These works provide adjacent candidate-mechanism comparators; none establishes those mechanisms in the present stack, and the configured-material protocol likewise treats such mechanisms as candidates rather than conclusions [P1].

Accordingly, the present construction borrows an experimental question from moiré and configured-material research - can relative spatial registration matter? - while withholding transfer of the microscopic explanation.

3. Layered carrier and configuration state

Let N deformable sheets be indexed by $i=1,\dots,N$. Each has a two-dimensional reference domain Σ_i and an embedding $\varphi_i:\Sigma_i\to\mathbb{R}^3$. For a radial specimen, use coordinates (r,ϑ) on an annulus $r\in[r_0,R]$. A general graph embedding is

$$\varphi_i(r,\vartheta) = (r\cos\vartheta,\ r\sin\vartheta,\ z_i(r,\vartheta)).$$

The configuration is not identified with external shape alone. Define

$$C = [\ \{\varphi_i\}, \{g_i,b_i\}, \{\chi_i\}, \{d_i\}, \{\mathscr{R}_i\}, \{G_{c,i}\}, \{S_i\}, p\ ].$$

Here g_i and b_i are the first and second fundamental forms; χ_i is an interlayer correspondence map; d_i is the signed normal-projected interlayer gap under the declared correspondence; $\mathscr{R}_i$ is an interlayer registry field; $G_{c,i}$ is a weighted contact structure; S_i is any controlled spacer field; and p contains process and environmental variables such as humidity, temperature, prestress, drying history, and material batch. The measured relation is simply

$$R : C \times E \to Y,$$

where E is a controlled excitation and Y is the measured observable vector. The purpose of the mathematics is to determine which compressed components of C are actually needed to predict Y.

4. Radial divot and frozen spatial modes

The proposed specimen uses a central divot or actuation region surrounded by prescribed radial corrugations. The divot provides a common geometric origin and a low-dimensional actuator; the surrounding ripple field provides a spatially encoded registry.

For a smooth radial fitting representation one may write

$$z_i(r;h) = D_i(r;h) + \Sigma_n A_{in} J_0(k_{in} (r-r_{in})),$$

Here the shifted J_0 term is used as a radial fitting basis; it is not asserted to be an axisymmetric harmonic solution about the divot center when $r_{in} \neq 0$ unless the declared coordinate origin and boundary conditions justify that interpretation. A central deformation term may be written, for example, as

$$D_i(r;h) = -h_i \exp[-r^2/(2\sigma_i^2)].$$

For fabrication and arithmetic analysis on an active annulus of length L=R-r0, a simpler periodic basis is often preferable:

$$z_n(r) = A \cos[2\pi n(r-r0)/L + \varphi], \quad \lambda_n = L/n.$$

The integer n therefore indexes a physically manufactured spatial division of one fixed carrier length. This is the precise sense in which an abstract divisor or harmonic index can be instantiated as a frozen surface-wave expression. The index remains an index; its physical meaning arises from the declared carrier and manufacturing map.

5. Geometry of a layer

For an axisymmetric graph $z_i=z_i(r)$, the induced metric is

$$g_i = \mathrm{diag}(1+(z_i')^2, r^2).$$

The principal curvatures are

$$\kappa_{r,i} = z_i'' / [1+(z_i')^2]^{(3/2)},$$

$$\kappa_{\vartheta,i} = z_i' / [r \, \mathrm{sqrt}(1+(z_i')^2)].$$

Hence

$$H_i = (\kappa_{r,i}+\kappa_{\vartheta,i})/2, \quad K_i = \kappa_{r,i} \kappa_{\vartheta,i}.$$

A smooth axisymmetric center requires $z_i'(0)=0$ when the center is included. These quantities are geometric observables or derived fields; none is yet an electrical mechanism.

6. Interlayer correspondence, separation, and mismatch

Two layers should not be compared merely by subtracting their plotted heights, because rotation, slip, lateral offset, and deformation can change which material points correspond. Define a partial interlayer correspondence

$$\chi_i : \Sigma_i \rightharpoonup \Sigma_{(i+1)}.$$

For a locally rigid in-plane transform,

$$\chi_i(x) = R(\Delta\theta_i)x + \Delta\delta_i,$$

and more generally $\Delta\theta_i$ and $\Delta\delta_i$ may be fields. Let $n_i(x)$ be the unit normal to layer i. The signed normal-projected interlayer gap under χ_i is

$$d_i(x) = [\varphi_{(i+1)}(\chi_i(x)) - \varphi_i(x)] \cdot n_i(x).$$

Define the radial component of the relative in-plane offset in the chosen radial chart by

$$\Delta r_i(x)=e_r(x)\cdot\Delta\delta_i(x).$$

The local registry state can then be written

$$\mathscr{R}_i(x) = (d_i(x), \Delta\theta_i(x), \Delta\delta_i(x), \Delta r_i(x)).$$

On every comparison patch U contained in $\mathrm{dom}(\chi_i)$, assume χ_i is at least C^1 where metric pullback is evaluated and sufficiently regular (taken here as C^2) where second-form comparison is evaluated. Slip lines, tears, or discontinuous correspondence boundaries are excluded from that smooth pullback and recorded separately.

Metric and curvature mismatch are defined by pulling the upper-layer geometry onto the lower layer:

$$\Delta g_i = \chi_i^* g_{(i+1)} - g_i, \quad \Delta b_i = \chi_i^* b_{(i+1)} - b_i.$$

This distinguishes intrinsic metric disagreement, curvature disagreement, separation, and lateral registry rather than collapsing all relative geometry into one projected image.

7. Registry superperiods and exact arithmetic structure

For two locally periodic modes

$$z_1(r)=A_1 \cos(k_1 r+\varphi_1), \quad z_2(r)=A_2 \cos(k_2 r+\varphi_2),$$

their relative phase is

$$\Phi(r)=(k_2-k_1)r+\Delta\varphi.$$

The fundamental recurrence of relative phase is therefore

$$L_{reg} = 2\pi/|k_2-k_1| = \lambda_1 \lambda_2 / |\lambda_1-\lambda_2|.$$

For regression and equal-mode limits, use the inverse registry-frequency descriptor $q_{reg} = 1/L_{reg} = |1/\lambda_1 - 1/\lambda_2| = |k_2-k_1|/(2\pi)$. Unlike L_{reg}, q_{reg} remains finite and tends continuously to 0 as $\lambda_1 \to \lambda_2$.

If $\lambda_2=\lambda(1+\varepsilon)$ with $|\varepsilon|<<1$, then $L_{reg} \approx \lambda/|\varepsilon|$. Small wavelength mismatch can therefore produce a much larger geometric registry scale.

For integer-indexed spatial modes $\lambda_n=L/n$ and $\lambda_m=L/m$ on one fixed active carrier,

$$L_{reg}(n,m) = L/|n-m|, \quad n\neq m.$$

Three separate arithmetic structures should be distinguished. First, $|n-m|$ controls the difference-mode or beat recurrence. Second, the common boundaries of the n-fold and m-fold equal partitions occur on a gcd(n,m)-fold partition of L. Third, a uniform q-fold partition refines both the n-fold and m-fold partitions exactly when n divides q and m divides q. The least such q is lcm(n,m); therefore the lcm partition is the coarsest common uniform refinement. Arbitrarily finer common refinements also exist. Thus

$$\text{difference scale} \leftrightarrow |n-m|; \quad \text{common partition} \leftrightarrow \gcd(n,m); \quad \text{coarsest common uniform refinement} \leftrightarrow \mathrm{lcm}(n,m).$$

These relations are exact for the constructed carrier. They do not imply that arithmetic alone determines the electrical response. They tell us which spatial coincidences the manufactured geometry makes available for testing.

8. Phase, amplitude, and predicted movement rules

A registry hypothesis becomes scientifically useful only if changing the geometry moves its predicted features before the response is observed. For fixed k_1 and k_2, equivalent relative-phase loci satisfy

$$r_j = [2\pi j-\Delta\varphi]/(k_2-k_1).$$

Changing phase translates the registry pattern while preserving L_{reg}. Changing amplitude A changes mismatch magnitude without changing the recurrence length. Changing λ_2 changes L_{reg} itself. These distinctions give three independent intervention rules:

$$\Delta\varphi \to \text{translation}; \quad A \to \text{mismatch strength}; \quad \Delta\lambda \to \text{recurrence scale}.$$

A response feature attributed to one of these variables must follow the corresponding movement rule. A feature that remains fixed when its proposed geometric driver is deliberately moved counts against that attribution.

9. Multimode and multilayer registry spectra

Real manufactured surfaces will contain multiple spatial components. Write

$$z_i(r)=\Sigma_n A_{in} \exp[i(k_{in} r+\varphi_{in})]$$

as a compact modal representation. Pairwise differences generate

$$\Delta k_{nm}^{(ij)}=k_{in}-k_{jm}, \quad L_{nm}^{(ij)}=2\pi/|\Delta k_{nm}^{(ij)}|.$$

Define a registry spectrum

$$\mathscr{S}_{ij} = \{ \Delta k_{nm}^{(ij)}, L_{nm}^{(ij)}, A_{in} A_{jm}, \Delta\varphi_{nm}^{(ij)} \}_{(n,m)}.$$

For three or more layers, the pairwise differences are constrained; for example $\Delta k_{13}=\Delta k_{12}+\Delta k_{23}$. A stack can therefore be manufactured with nested registry scales. Joint alignment events may occur less frequently than pairwise events, creating a hierarchy of predicted configuration loci without any assumption that the layers support coherent quantum waves.

10. Contact as a weighted field and topology as a separate claim

Let local effective interface conductance or coupling be represented by an initially unspecified function

$$g_i(x)=\mathscr{G}[d_i(x),P_i(x),\mathscr{R}_i(x),p].$$

The function is intentionally left empirical at the first stage. For a declared threshold τ, define an active-contact set

$$C_i(\tau)=\{x\in\Sigma_i : g_i(x)\geq\tau\}.$$

Geometry may change g_i continuously while the topology of C_i remains unchanged. A topological transition is reserved for changes in connectivity descriptors such as Betti numbers β_0 and β_1 or their persistent-threshold analogues. This preserves the distinction already required by the configured-material protocol [P1].

The experimentally stronger question is whether response is better predicted by contact area alone, by contact-network connectivity, or by registry variables after either contact descriptor is included.

11. Configuration-space traversal by central displacement

Let h denote controlled central displacement. Then the material traverses a path

$$\gamma : h \mapsto C(h) = \{\varphi_i(h),\chi_i(h),d_i(h),G_{c,i}(h),\ldots\}.$$

The measured response is

$$Y(h)=R[C(h),E(h)].$$

Loading and unloading define γ_{up} and γ_{down}. Response hysteresis can be summarized by

$$\mathscr{H}_Y = \oint Y\, dh.$$

A nonzero loop indicates path dependence but does not identify its mechanism. Friction, adhesion, viscoelasticity, humidity, retained contact, snap-through, or another process must be discriminated separately.

12. Spatial compression as an identifiability test

The full field C may contain far more information than is required to predict Y. Let S(C) be a lower-dimensional descriptor. Candidate levels include

$$S_0=(\varepsilon,\kappa),$$

$$S_1=S_0+(d,\text{contact metrics}),$$

$$S_2=S_1+(\hat{q}_{reg},\Delta\hat{\varphi}_*,I_0,\bar{n},\text{gcd/lcm nominal design descriptors},\text{selected registry-spectrum terms}),$$

$$S_3=S_2+(\beta_0,\beta_1,\text{persistent-contact summaries}).$$

Compression is adequate for the measured task if retaining the full configuration does not materially improve prediction after S(C) and the excitation are known. Operationally, the compression fails if two configurations have the same proposed summary but produce distinguishable responses beyond declared uncertainty.

$$d_S(\hat{S}(C_1),\hat{S}(C_2)) \le \varepsilon_S \text{ and } d_Y(Y(C_1,E),Y(C_2,E)) > \varepsilon_Y \;\Rightarrow\; S \text{ is inadequate for that response task.}$$

Here d_S and d_Y are preregistered comparison metrics and ε_S, ε_Y are tolerances chosen from measurement uncertainty or an externally declared practical resolution. Exact equality of experimentally estimated summaries is not required. This turns 'compression of relations in space' into an experimentally falsifiable proposition rather than a metaphor.

13. Model ladder

The first-stage analysis uses a nested predictive sufficiency ladder. It is not a causal partition. Registry geometry may physically alter realized contact, so information already captured by measured contact descriptors is not counted again as independent registry information. If M1 captures that mediation completely, M2 should add no held-out predictive value; that outcome supports M1 sufficiency rather than implying that geometry had no physical influence.

M0 - Bulk deformation/environment:

$$M_0:\ Y=F_0(\varepsilon,\kappa,h,\dot{h},T,RH,P,\ldots).$$

M1 - Contact/separation/network:

$$M_1:\ Y=F_1(C_{bulk},G_c,d,P_c,C_{env}).$$

M2 - Registry geometry:

$$M_2:\ Y=F_2(C_{bulk},C_{contact},C_{reg},C_{env}).$$

M3 - Material-specific interlayer structure:

$$M_3:\ Y=F_3(C_{bulk},C_{contact},C_{reg},C_{mat},C_{env}).$$

M4,j - Mechanism-specific branches: percolation/tunneling, piezo/flexoelectric coupling, ionic/hydration redistribution, molecular charge transfer, electronic interference/band-structure effects, or other independently testable mechanisms.

The first four levels are approximately nested by added predictive variables; the M4,j mechanisms need not be mutually exclusive. A later model is not declared 'more real.' It earns standing only when the added structure buys reproducible predictive information or unique discriminating consequences.

14. Promotion and retirement rules

A more complex model should be promoted only if it improves held-out prediction by a preregistered practically meaningful margin and survives new specimens, new geometry, randomized measurement order, and direct transformation tests. Prediction error must be evaluated on held-out specimens rather than held-out cycles from the same specimen.

$$\text{Promote } M_{j+1} \text{ only if } \mathscr{L}_{test}^{bal}(M_{j+1}) < \mathscr{L}_{test}^{bal}(M_j) - \Delta_{min},$$

with Δ_min fixed before model comparison. Complexity penalties or cross-validation should prevent parameter-rich models from receiving credit merely for fitting training data.

A model is retired only as a sufficient description within the tested domain. M0 becomes insufficient if registry-different states with matched bulk deformation remain distinguishable. M1 becomes insufficient if registry variables add held-out predictive value after contact metrics are controlled. M2 fails to earn standing if registry variables do not improve held-out prediction or if preregistered migration rules fail. A specific M4,j loses standing if its unique discriminating prediction fails.

15. Controlled discriminating experiment

The primary experiment uses an annular active region r∈[r0,R] with L=R-r0 and a controlled central actuator. The first geometry screen fixes chemistry, nominal amplitude, electrode geometry, environment, and fabrication batch while varying only spatial mode pair and relative phase.

The first screen uses six modulated geometry classes plus an unmodulated matched-roughness control. The added E=(7,9) class supplies an additional arithmetic contrast to the original B/C comparison, while the U control separates effects of prescribed periodic structure from effects of merely having a structured or roughened interface. U is fabricated without a periodic radial mode pair and is matched to the modulated coupons on preregistered nonperiodic surface statistics - primary RMS height over the declared spatial band, with amplitude range and roughness spectrum reported as secondary matching diagnostics:

Class	Modes	Phase	qreg L	Purpose / arithmetic control
A0	(6,6)	0	0	equal-mode aligned control; gcd=6
Aπ	(6,6)	π	0	phase-only control; gcd=6
B	(6,7)	fixed	1	one slow cycle; gcd=1; lcm=42
C	(6,8)	fixed	2	two slow cycles; gcd=2; lcm=24
D	(9,10)	fixed	1	same difference scale as B; gcd=1; lcm=90
E	(7,9)	fixed	2	same difference scale as C; gcd=1; lcm=63
U	unmodulated	n/a	n/a	matched-roughness / corrugation-null control

Each class is crossed with two controlled applied-preload conditions, producing 14 fabricated experimental states. An initial allocation of three to five independent stacks per state therefore requires 42-70 independent assemblies. This is a pilot/variance-estimation burden, not a declaration of confirmatory power; the final confirmatory independent-specimen count is fixed only after the pilot/precision calculation in Section 17.1.

Applied preload P is an intervention, not a complete observation of realized contact. Realized contact descriptors - including measured or estimated separation, true/contact-proxy area, and contact-network summaries where available - are recorded separately for M1. Repeated cycles are nested measurements and must not be counted as independent specimens. Tooling is part of the fabrication hierarchy. The confirmatory design should use at least two independently fabricated molds or masters for every modulated mode-pair class where the manufacturing route permits, distribute specimens across those tools, and record mold/master identity as a crossed fabrication factor. The U control used for the Section 16 structured-surface contrast should likewise be produced from at least two independently fabricated roughness-control tools or independent fabrication masters where feasible. If only one mold/master is used for a mode-pair class, mode-pair identity and tooling remain aliased; if only one U tool/master is used, the structured-surface contrast remains tooling-aliased. In either case the reported claim must retain the corresponding bound.

The primary response channels are through-interface resistance or conductance R_perp(h), in-plane resistance R_parallel(h), high-impedance/open-circuit voltage V_oc(t), and if feasible applied force F(h). Temperature and humidity are continuously logged. The existing Part I protocol's artifact controls - cable motion, electrostatic pickup, triboelectric contact, contact resistance, instrument loading, cracking, batch gradients, and return-to-state behavior - transfer directly to this experiment [P1].

16. Primary predictions and blind tests

The first strong test is movement, not amplitude. Configuration B=(6,7) has q_reg L=1. Configuration C=(6,8) has q_reg L=2. Because B→C simultaneously changes |n-m| and gcd(n,m), that contrast alone cannot identify which arithmetic descriptor carries the predictive information. Configuration E=(7,9) adds a further contrast: C and E share the same dimensionless difference-scale value q_reg L=2, while gcd(6,8)=2 and gcd(7,9)=1. However, C→E also changes mean mode index from 7.0 to 8.0 and lcm from 24 to 63. It is therefore not a pure gcd intervention. C/E is read jointly with B/D, but no monotonic transfer is assumed: a null B/D result constrains carrier- and refinement-associated explanations only over the B/D contrast; it does not establish their absence at C/E and therefore does not convert C/E into a pure gcd test. Movement attributed specifically to the difference scale must survive these controls and follow the preregistered transformation under the measured manufactured geometry.

The second blind test compares B=(6,7) with D=(9,10). Both have |n-m|=1, q_reg L=1, and gcd=1, while their mean mode index changes from 6.5 to 9.5, a ratio of about 1.46, and lcm changes from 42 to 90. B/D is therefore a joint carrier-and-refinement contrast, not a pure carrier intervention. A null result constrains sensitivity to those jointly changed descriptors over this tested contrast; a positive result shows that the slow difference scale alone is insufficient but does not identify whether carrier scale, lcm/refinement structure, or their covariation is responsible.

The phase-only test compares A0=(6,6,0) and Aπ=(6,6,π). In the ideal equal-mode design they have q_reg=0 and differ in a global crest-registry phase. For unequal-mode classes B/C/D/E, relative phase varies with radius. Define the reference radius by the analyzed annulus, r_*=(r_min+r_max)/2, fixed before response inspection, and let Δφ̂_* denote the fitted relative phase at r_*. Cross-class comparisons of Δφ̂_* are conditional on this reference-location convention; r_* need not represent the same fraction of a registry cycle in different classes. An equal-mode-design indicator I_0 marks A0/Aπ. In X2 the two phase regimes enter with separate coefficients through I_0 Δφ̂_* and (1-I_0) Δφ̂_*, rather than through one shared phase slope. The unmodulated matched-roughness control U supplies a separate comparison: it asks whether prescribed periodic registry contributes beyond a structured interface matched primarily on RMS height over the preregistered spatial band, with amplitude range and roughness spectrum retained as matching diagnostics.

A useful intervention crosses low/high registry alignment with low/high applied preload. The interaction statistic

$$I_{(reg\times P)} = (Y_{highReg,highP} - Y_{lowReg,highP}) - (Y_{highReg,lowP} - Y_{lowReg,lowP})$$

measures a preload-by-registry contrast. It does not by itself establish a contact-independent registry effect, because registry can change the realized contact field even at fixed preload. Promotion from M1 to M2 therefore requires held-out predictive improvement after the measured or estimated realized-contact descriptors are included.

17. Statistical and predictive analysis

For every physical specimen, the manufactured geometry should be measured rather than assumed from the mold. Estimated wavelengths, amplitudes, phase, divot shape, and any measurable separation/contact descriptors receive hats and become the actual predictors. In particular,

$\hat{q}_{reg} = |1/\hat{\lambda}_1 - 1/\hat{\lambda}_2|$. For every modulated specimen, including nominal A0/Aπ specimens, $\hat{q}_{reg}$ is computed from the measured fitted wavelengths and is not forced to zero. The value $q_{reg}=0$ denotes only the ideal equal-mode limit. The primary X2 model assumes that response may track realized $\hat{q}_{reg}$ regardless of whether that realized variation was deliberately designed (B-E) or arose as fabrication deviation around the equal-mode A design. That provenance-independence is an explicit modeling assumption rather than an identity; I_0 is retained so equal-mode-design specimens remain distinguishable. For unequal-mode specimens, $\Delta\hat{\varphi}_*$ is evaluated at $r_*=(r_{min}+r_{max})/2$; for the A design the phase term represents global crest registry. The two phase regimes receive separate coefficients as specified above. By contrast, gcd(n,m), lcm(n,m), and the nominal mean mode index $\bar{n}=(n+m)/2$ are descriptors of the prescribed integer mode pair, not quantities recovered by rounding measured wavelengths to exact rational ratios. They therefore carry no hats and are labeled nominal design descriptors wherever they enter X2.

A mixed or hierarchical model should treat cycles and displacement samples as nested within independent specimens. One schematic full-factor training model, shown only to illustrate specimen nesting and fixed design factors rather than to define X0, is

$$Y_{ijkl} = \beta_0 + \beta_R R_i + \beta_P P_j + \beta_{RP} R_i P_j + b_k + \varepsilon_{ijkl},$$

where registry class R_i and applied-preload condition P_j are fixed effects and b_k is a specimen-level random effect. Applied preload P is present already in M0 as a manipulated design variable, while realized-contact descriptors enter M1 separately. The displayed R_i and $R_i \times P_j$ terms illustrate a registry-inclusive training form and therefore should not be read as the M0 predictor set. The random effect captures within-training-specimen heterogeneity; it is not estimated from the held-out specimen when scoring R4.

The central model-comparison analysis builds nested predictor sets X0, X1, and X2 corresponding to M0, M1, and M2, then evaluates held-out-specimen prediction error on one common population: the 12 modulated geometry-by-preload states. X2 uses measured $\hat{q}_{reg}$; the separate phase-regime terms $I_0\, \Delta\hat{\varphi}_*$ and $(1-I_0)\, \Delta\hat{\varphi}_*$; the equal-mode-design indicator I_0; measured carrier/registry-spectrum descriptors where defined; and explicitly nominal design descriptors such as $\bar{n}$, gcd, and lcm. Because ordinary specimen-level folds contain training specimens from the same mode pairs as the test specimens, predictive gain from nominal arithmetic terms can behave partly as a class-level signal and does not by itself demonstrate transfer to unseen geometry. Accordingly, the primary R4 score remains specimen-held-out generalization within the preregistered tested mode pairs, and a secondary leave-one-mode-pair-out analysis is preregistered to test transfer to an unseen mode pair. A0 and Aπ are held out together because both instantiate the (6,6) mode pair; both preload states associated with the held-out pair are also excluded from training. Failure of that secondary test does not erase within-pair R4 performance, but it blocks interpreting nominal arithmetic descriptors as a demonstrated cross-mode-pair law. The two unmodulated U states are excluded from the X0/X1/X2 loss comparison entirely and are used only for the preregistered structured-surface contrast in Section 16 because no mode pair, phase, or registry descriptor is defined for U. Registry earns support only if M2 improves held-out performance by the preregistered margin and the blind movement tests survive.

17.1 Primary endpoint and preregistration rule

For the confirmatory M0/M1/M2 comparison, the primary electrical target is the through-interface resistance trajectory over a preregistered displacement domain. For each specimen s, valid positive resistance measurements are represented as

$$y_s(h) = \ln R_{\perp,s}(h),$$

interpolated only onto a preregistered common displacement grid inside the range actually observed for that specimen. No extrapolated points enter the primary loss. If the instrumentation or specimen makes positive finite R_{perp} unavailable over the declared domain, that specimen is reported as endpoint-ineligible under a preregistered missingness rule rather than silently transformed into another endpoint. For each experimental state g, endpoint eligibility is reported as $e_g=N_{eligible,g}/N_{attempted,g}$, and achieved displacement-domain extent is reported as $D_g=h_{max,g}-h_{min,g}$. Differential eligibility or domain truncation is treated as an experimental outcome rather than discarded as nuisance missingness. The primary model comparison is accompanied by a preregistered sensitivity analysis on a common domain supported across the compared states.

The confirmatory primary test loss is state-balanced trajectory root-mean-square error; the specimen-unweighted mean is retained as a sensitivity result.

$$L_{test}^{bal}(M) = (1/|G_{req}|) \sum_{g\in G_{req}} [(1/|S_{g,test}|) \sum_{s\in S_{g,test}} RMSE_s(M)], \quad RMSE_s(M) = \sqrt{(1/|G_s|) \sum_{h\in G_s} (\hat{y}_{M,s}(h)-y_s(h))^2}. \quad L_{test}^{unw}(M) = (1/|S_{test}|) \sum_{s\in S_{test}} RMSE_s(M).$$

Here G_{req} is the preregistered set of modulated geometry-by-preload states required in the test fold, and $S_{g,test}$ is the eligible held-out specimen set for state g. If any required state has $|S_{g,test}|=0$, L_{test}^{bal} for that fold is undefined: the analysis does not drop the state and renormalize the remaining states. The fold is reported non-evaluable for confirmatory R4 (or the preregistered grouped resampling scheme is used only if it can restore at least one eligible held-out specimen per required state without using response information). Both L_{test}^{bal} and L_{test}^{unw}, together with e_g and D_g, are reported.

All displacement samples and repeated cycles from one physical stack remain in the same fold. Folds are grouped at the physical-specimen level and stratified by geometry-by-preload state as far as the final sample permits. For a held-out specimen, prediction is population-level: its specimen-specific random effect is fixed at $b_k=0$ (equivalently, no cycles or observations from that specimen may be used to estimate or calibrate b_k before scoring). A separately labeled within-specimen calibration analysis may be reported exploratorily, but it cannot contribute to R4. Fold construction, the required state set G_{req}, and any rule for handling a non-evaluable balanced fold are frozen before confirmatory unblinding.

The practical improvement threshold Δ_{min} is expressed in the same units as the primary loss - natural-log-resistance RMSE - and is fixed before confirmatory unblinding. It is chosen from a separately identified pilot/noise set or from an externally declared engineering tolerance and is then frozen with the analysis script and configuration table. It may not be selected from the confirmatory model-comparison results. The power/precision estimand is the state-balanced held-out loss improvement $\Delta L_{test}^{bal}=L_{test}^{bal}(M1)-L_{test}^{bal}(M2)$, not an individual fixed-effect coefficient. The confirmatory independent-specimen count is therefore fixed before unblinding from a preregistered calculation using pilot estimates of the uncertainty of fold- or resampling-paired ΔL_{test}^{bal} under the same grouped evaluation procedure, together with the declared Δ_{min} and target operating criterion. Until that calculation is performed, $n=3$ or $n=5$ per state is a pilot allocation, not evidence of adequate confirmatory power.

Blind movement tests and predictive improvement remain distinct, non-nested endpoints. Passing a movement test supports R3; it is not a prerequisite logically implied by R4. Defeating M1 by the preregistered held-out margin supports R4; it does not imply that a particular movement test passed. The evidence table orders increasing kinds of warrant for convenience, not a strict logical implication chain. The paper's strongest initial positive statement requires both: at least one preregistered registry transformation must move the response structure in the predicted direction, and M2 must improve held-out prediction over M1 by Δ_{min}. Secondary channels such as $R_{parallel}$, V_{oc}, force, hysteresis, endpoint eligibility, achieved-domain extent, and contact-network summaries remain diagnostic unless separately preregistered as additional confirmatory endpoints.

18. Evidence ladder and maximum initial claim

The paper uses a registry-specific evidence ladder:

Level	Requirement	Maximum warranted statement
R0	manufactured geometry measured	declared registry field exists geometrically
R1	reproducible response	specified configuration changes measured response
R2	registry covariation	response covaries with registry descriptor
R3	blind transformation prediction	changing registry moves response as preregistered
R4	simpler models defeated	registry adds held-out prediction beyond bulk/contact
R5	material substitution	material identity adds structured contribution
R6	mechanism discrimination	one microscopic model survives unique tests better than alternatives

The strongest initial positive statement is intentionally bounded: within the tested material system, deliberately prescribed interlayer spatial registry supplies reproducible predictive information about the measured response beyond the registered bulk-deformation and measured/estimated realized-contact variables, and altering the registry geometry moves response structure in the preregistered direction. The experiment does not by itself identify the microscopic mechanism responsible for that coupling.

19. Negative outcomes that remain scientifically useful

The experiment is designed so that failure contracts the claim rather than destroying the research program. No reproducible electrical response means the implementation fails at the response level. A reproducible response explained by bulk deformation leaves M0 sufficient. Internal-geometry dependence explained by contact leaves M1 sufficient. Training-data registry correlation without blind migration implies overfitting or missing variables. Batch-specific registry effects indicate that manufacturing/process variables dominate the present claim. These are informative outcomes because each localizes where additional structure did or did not earn predictive standing.

This follows the larger open-replication program's explicit evidence ladder and its requirement that matched controls, artifact checks, independent replication, and prediction determine promotion [P1-P3].

20. Material-specific and fullerene branches

C60, graphene substitution, or other interlayer materials are deliberately excluded from the primary M0/M1/M2 discrimination. If a spacer is mechanically necessary, the first experiment should use the most inert and reproducible practical choice. Material identity becomes a controlled variable only after the geometry experiment has either supported or failed M2.

In a second-stage material substitution, geometry is held as nearly fixed as practical while spacer or conductor identity is changed. If feature location in configuration space remains stable while amplitude changes, geometry may set the locus while material controls coupling strength. If the feature disappears, material-specific coupling is required. If the feature moves, an absolute material length or interaction scale participates in the locus.

Only after such results should specific microscopic models - including molecular charge transfer or genuine electronic interference - receive dedicated experiments. Resemblance to fullerene/graphene or moiré literature is not itself evidence that the same mechanism operates here.

21. Novelty boundary and targeted prior-art collision search

The paper does not claim novelty for layered materials, corrugated conductive films, strain sensing, moiré structures, pressure/twist tuning, contact percolation, chitosan composites, radial or circular gratings, or topological data analysis individually. Each has substantial prior art. A targeted prior-art collision search was rerun on 22 August 2026 for this rebuild. Exact query families included: "radial corrugation layered conductive material interlayer registry"; "circular radial moire grating displacement strain"; "concentric circle moire grating"; "concentric spiral few-layer graphene"; "interlayer registry corrugation twisted bilayer graphene"; "chitosan graphene radial corrugation registry"; "central actuator interlayer registry layered material"; and "gcd lcm spatial modes material registry". Sources examined through the search interface included publisher/index pages from Elsevier/ScienceDirect, Optica, ACS Publications, arXiv, PubMed where surfaced, and Google Patents; DOI records were followed where available. The bounded literature record retained for citation located established circular/radial moiré gratings for displacement and strain measurement [8], earlier concentric-circle moiré constructions [9], concentric and spiral few-layer graphene structures [10], and registry-dependent corrugation/interlayer energetics in twisted bilayer graphene [11]. Patent searching formed part of the collision screen, but no uncited patent-family result is relied upon in the contribution statement. The dated query list and source classes define the bound of this search; they do not make it exhaustive.

The searched literature therefore narrows the candidate contribution rather than proving novelty. The candidate contribution remains the integrated experimental object: a low-cost layered conductive or conductive-composite material with deliberately manufactured radial spatial modes treated as a traversable interlayer registry field; an exact arithmetic/geometric accounting of difference, common partition, and common refinement scales on a fixed carrier; a controlled central actuator that scans that manufactured field; and a preregistered model-selection protocol designed to determine whether measured registry descriptors add held-out predictive information beyond bulk strain and contact. The targeted search did not identify, within the queries and sources examined, a publication combining that full experimental architecture. This is a bounded search result, not a patentability opinion, exhaustive prior-art determination, or claim of invention.

Accordingly, any novelty or priority language remains conditional. The recorded search is bounded by its stated date, query families, and source classes and is reproducible as a search protocol, not exhaustive as a novelty determination. Before a formal priority, patentability, freedom-to-operate, or exhaustive novelty assertion, the search should be extended through specialist bibliographic and patent databases using the final claims and fabrication details. For this research note, the contribution is described only as a candidate integrated research contribution whose nearest located components and collision risks are disclosed.

22. Reproducibility package and future commercially sourced BOM

The eventual publication package should include: CAD or dimensioned drawings of the radial molds; a configuration table for every specimen; measured surface-profile parameters; electrode geometry; actuator geometry; raw electrical/mechanical waveforms; temperature and humidity logs; randomization scheme; analysis code; preregistered model-comparison thresholds; null and positive results; and a complete commercially sourced bill of materials with substitutions and supplier traceability.

The BOM is intentionally deferred until the mathematics and discriminating experiment have fixed the specifications. Commercial sourcing should therefore implement the experiment rather than silently determine it.

23. Limitations

The first-generation experiment cannot independently resolve nanometre-scale interlayer separation, atomic registry, local band structure, molecular orientation, or quantum coherence. Garage-scale profilometry and electrical measurements can test macroscopic registry-response relations, not atomic moiré physics.

The radial-wave basis is an idealized descriptor of manufactured surfaces. Real chitosan/composite sheets will contain roughness, thickness variation, anisotropy, drying stress, defects, humidity gradients, and local slip. The measured geometry rather than CAD intent must therefore enter the predictive model. The U corrugation-null is itself fabrication-sensitive: matching a nonperiodic control to a modulated surface on RMS height over a declared spatial band while suppressing periodic registry may be difficult. For a sinusoidal target of amplitude A the corresponding RMS height is approximately A/sqrt(2), so the U surface can remain substantially rough. Failure to achieve the preregistered roughness match weakens or invalidates that particular U contrast and must be reported rather than absorbed into the registry interpretation.

Contact and registry may be strongly coupled, making perfect orthogonal control difficult. The proposed preload factorial, measured-contact covariates, and degeneracy tests reduce this problem but do not guarantee complete identifiability. A result should be reported at the strongest level actually supported by the controls.

Finally, a successful registry relation would establish a new descriptive and engineering handle before it establishes a microscopic mechanism. That ordering is intentional.

24. Conclusion

A configured multilayer material can be treated as more than a stack of bulk constitutive sheets. Relative spatial registration can itself be manufactured, measured, transformed, and tested as a predictive variable. The radial-divot architecture makes that proposition unusually accessible: a common center anchors the layers; integer-indexed spatial modes encode controlled divisions of a fixed carrier; their differences generate calculable superperiods; their gcd and lcm structures separate common partition from coarsest common uniform refinement; and one macroscopic displacement can sweep the stack through a high-dimensional internal registry landscape.

The proposed experiment does not begin by assuming that the material will display moiré electronics, magic-angle behavior, topological bands, unusual charge transfer, or any other exotic mechanism. It begins with a harder and more useful question: does the deliberately constructed relation predict anything beyond bulk deformation and contact? If not, the model contracts. If yes, the geometry has earned further investigation. If the feature then moves with preregistered registry transformations, survives matched controls, and predicts blind configurations, the resulting phenomenon is scientifically substantive even before its microscopic explanation is settled.

This sequencing - construct, calculate, perturb, predict, measure, discriminate, and only then interpret - is the central methodological contribution of the paper.

Appendix A. Bounded audit closure record

The publication edition incorporates the seven corrections frozen at the end of the Step-7 bounded adversarial audit, the subsequent bounded R1-R4 correction sets, the R5 external-reference/prior-art verification pass, and this rebuilt R6 identifiability/inspectability correction set. Appendix A remains a revision ledger: earlier dispositions remain visible when later work supersedes their interpretation, and supersession is marked rather than silently rewritten.

Audit item	Publication disposition
A-1 lcm refinement direction	Corrected: lcm(n,m) is identified as the coarsest common uniform refinement, with the divisibility condition stated explicitly.
A-2 preload versus realized contact	Corrected: applied preload P is separated from realized-contact descriptors; the interaction is interpreted only as a preload-by-registry contrast.
A-3 primary predictive endpoint	Corrected: Section 17.1 freezes the primary trajectory target, grouped specimen-level validation, test loss, Δ_min rule, and relation between R3 and R4.
B-1 meaning of d_i	Corrected: d_i is a signed normal-projected correspondence gap, not a general minimum surface separation.
B-2 undefined Δr_i	Corrected: Δr_i is defined as the radial component $e\_r \cdot \Delta\delta\_i$.
B-3 regularity of χ_i	Corrected: smoothness/domain conditions are stated for metric and second-form pullbacks; discontinuity boundaries are treated separately.
B-4 model-ladder interpretation	Corrected: M0/M1/M2 is explicitly a predictive sufficiency ladder rather than a causal partition.
R1-1 arithmetic-design confounding	Corrected: E=(7,9) holds |n-m|=2 and Lreg=L/2 while changing gcd from 2 to 1 relative to C=(6,8). SUPERSEDED BY R3-2 (and clarified in R4): retained as historical closure record; later revisions reject interpretation of C/E as an isolated gcd contrast.
R1-2 held-out random effects	Corrected: R4 uses population-level prediction for a held-out specimen with $b\_k=0$ and no within-specimen calibration.
R1-3 endpoint eligibility	Corrected: eligibility rate and achieved displacement-domain extent are reported by state with a common-domain sensitivity analysis.
R1-4 power / Δmin	Corrected: Δmin units are ln-resistance RMSE; confirmatory n is determined from preregistered pilot-based power/precision, not assumed from n=5.
R1-5 precision controls	Corrected: P is explicit in M0; compression uses tolerance metrics; B/D contrast is quantified; U control added; R3/R4 declared non-nested.
R2-1 abstract/state count	Corrected: abstract now states seven classes crossed with two preload conditions, giving 14 primary states.
R2-2 C/E interpretation	Corrected: C/E is not called a pure gcd contrast;

	mean-mode and lcm changes are disclosed and interpretation is joint with B/D.
R2-3 equal-mode registry predictor	Corrected: X2 uses q_reg=1/L_reg, finite with equal-mode limit q_reg=0, rather than noise-dominated L_reg. SUPERSEDED BY R3-4: retained as historical closure record; q_reg=0 denotes the ideal equal-mode limit, not an assigned measured value for nominal A specimens.
R2-4 U control entry rule	Corrected: U is excluded from continuous X0/X1/X2 loss fits and reserved for the structured-surface contrast; roughness matching is defined.
R2-5 notation/model/power residue	Corrected in R2: e_g and D_g avoid endpoint-symbol overload; G_s is the RMSE grid; the training model is labeled registry-inclusive; measured Y is unhatted in §12. SUPERSEDED IN PART BY R6-3/R6-7: the power estimand is now ΔL_test^bal and the required balanced-loss state set is denoted G_req.
R3-1 state-population scope	Corrected: 14 fabricated states are distinguished from the 12 modulated states used for the common M0/M1/M2 loss comparison; U is explicitly excluded from that comparison.
R3-2 B/D and C/E attribution	Corrected: B/D is disclosed as a joint carrier-and-lcm contrast; C/E remains multiply attributable; no monotonic transfer from a B/D null is assumed.
R3-3 phase commensurability	Corrected: unequal-mode phase is Δφ_* at preregistered r_* and an equal-mode indicator I_0 separates the A-class global phase relation.
R3-4 measured qreg	Corrected: qhat_reg is computed from measured fitted wavelengths for all modulated specimens, including nominal A classes; zero is only the ideal equal-mode limit.
R3-5 design burden/control risk	Corrected: table uses qreg L; 42-70 pilot assemblies are stated; U roughness matching is registered as a fabrication-sensitive control limitation.
R4-1 audit-ledger supersession	Corrected: R1-1 and R2-3 remain verbatim as history but are explicitly marked superseded by their later corrections.
R4-2 phase-regime coefficients	Corrected: X2 uses separate phase slopes I_0 Δφ̂_* and (1-I_0) Δφ̂_* rather than one shared coefficient.
R4-3 reference-radius rule	Corrected: r_*=(r_min+r_max)/2 is fixed from the analyzed annulus before response inspection; cross-class phase comparison is conditional on that convention.
R4-4 q̂_reg provenance assumption	Corrected: pooling realized q̂_reg across designed

	and fabrication-deviation regimes is declared as a modeling assumption, with I_0 retaining design provenance.
R4-5 design-table completeness	Corrected: lcm values are listed for B=42, C=24, D=90, E=63; redundant |n-m|/q_reg L wording removed.
R4-6 experiment title/burden	Corrected: Section 15 is titled Controlled discriminating experiment; the 42-70 assembly pilot burden remains explicit.
R5-1 references [1]-[6]	Verified against publisher/index records: titles, venues, years, volume/page or article identifiers, and DOIs match the cited records. The Section 2 attribution language was tightened to match what [4]-[6] directly support.
R5-2 reference [7] publication status	Corrected: the former arXiv-only citation is now cited to the peer-reviewed 2026 Advanced Materials article, DOI 10.1002/adma.202518224, while retaining the earlier arXiv identifier as provenance.
R5-3 targeted prior-art search	Performed and recorded in Section 21. Located adjacent prior art in circular/radial moiré metrology, concentric moiré grids, concentric/spiral few-layer graphene, and registry-dependent corrugation. R6 makes the search date, query families, source classes, and inclusion of patent searching explicit. No exhaustive novelty or patentability conclusion is asserted.
R5-4 novelty language boundary	Corrected: Section 21 now distinguishes a bounded search result from a novelty, priority, invention, patentability, or freedom-to-operate claim.
R6-1 measured versus nominal arithmetic descriptors	Corrected: q̂_reg and Δφ̂_* remain measured predictors; n̄, gcd, and lcm are explicitly nominal descriptors of the prescribed integer mode pair and are not inferred by rationalizing measured wavelengths. A secondary leave-one-mode-pair-out analysis tests transfer to an unseen mode pair, with A0/Aπ held out together for (6,6).
R6-2 tooling alias	Corrected: confirmatory modulated classes use at least two independently fabricated molds/masters where feasible, with tooling identity recorded; a one-tool class retains an explicit R4 alias bound. The U structured-surface control is now subject to the same independent-tooling principle, with a corresponding §16 bound if only one U tool/master is used.
R6-3 state-balanced confirmatory loss	Corrected: L_test^bal is the confirmatory loss, L_test^unw is retained as sensitivity, ΔL_test^bal is

	the power/precision estimand, and §14 uses the balanced loss in the promotion rule. A required state with zero eligible held-out specimens makes the fold non-evaluable rather than silently renormalized.
R6-4 prior-art search inspectability	Corrected: Section 21 records the 22 August 2026 search date, exact query families, source classes including Google Patents, DOI-following practice, and boundedness. Only cited literature [8]-[11] is relied upon in the contribution statement; no unidentified patent family is carried as evidence.
R6-5 shifted J0 status	Corrected: the shifted J_0 expression is identified as a radial fitting basis and is not asserted to be an axisymmetric harmonic solution about the divot center for r_in≠0 absent coordinate and boundary conditions that justify that interpretation.
R6-6 reference [10] authors	Corrected: the Chemistry of Materials reference [10] now includes its author list.
R6-7 notation harmonization	Corrected: S_2 uses measured q̂_reg and Δφ̂_* consistently with §17; the nominal mean-mode descriptor is n̄ throughout; and the required balanced-loss state set is G_req, avoiding collision with sinusoid amplitude A and the A-class label.

The closure rule for this edition is deliberately bounded: only the corrected passages and their direct dependency neighborhoods are rechecked for consistency. No new mechanism claim is introduced as part of closure.

Appendix B. Confirmatory analysis record template

A confirmatory replication should archive, before unblinding, at minimum: specimen IDs and state assignments; geometry measurements and fitted mode parameters, including qhat_reg, the preregistered phase reference radius r_*, and the equal-mode indicator; nominal arithmetic design descriptors kept distinct from measured predictors; applied-preload levels; realized-contact descriptors; mold/master IDs and independent-tooling assignments for modulated classes and the U control; common displacement grid; endpoint eligibility rule; statewise eligibility and achieved-domain summaries; grouped fold assignment; required balanced-loss state set G_req and non-evaluable-fold rule; explicit held-out random-effect rule (b_k=0); model formulas or code commit; L_test^bal as the confirmatory loss and L_test^unw as sensitivity; the secondary leave-one-mode-pair-out specification; Δ_min in natural-log-resistance RMSE units and its source; pilot variance estimates; the preregistered power/precision calculation fixing independent-specimen count; common-domain sensitivity analysis; U-control RMS-height band and matching diagnostics; blind movement predictions; environmental limits; exclusion rules; the dated prior-art query/source record; and hashes of the frozen analysis/configuration files. This record does not certify a physical mechanism. It makes the declared prediction and analysis boundary inspectable.

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